

# SPARROW Bill of Materials

Hardware sourcing checklist for the SPARROW Assembly and Setup Guide

Source basis: SPARROW Setup Guide Pi.docx, V1.2

## Purpose

This BOM is intended to sit beside the SPARROW assembly guide in the GitHub repository so builders can source the parts needed to assemble a standard SPARROW unit and identify optional parts for tougher field deployments or added sensors.

## Build assumptions

- Standard build means a Raspberry Pi 5 based SPARROW unit using solar power, a 24V battery bank, Starlink Mini networking, XBee RF support, Relay 3 Click power switching, and a weatherproof main enclosure.
- Optional items are included because the guide supports them, but they are not required for every deployment. These include AudioMoth, Environment 3 Click, Reolink cameras, tougher junction boxes, extra glands, and site-specific mounting hardware.
- Quantities are minimum recommended quantities for one SPARROW unit unless marked as site dependent, as needed, or per sensor.

## Core BOM for one standard SPARROW unit

| System              | Component                                        | Qty | Status             | Use in SPARROW                                                                                                          | Sourcing notes                                                                                |
|---------------------|--------------------------------------------------|-----|--------------------|-------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Compute and storage | Raspberry Pi 5, 8GB RAM                          | 1   | Required           | Main edge computing device. Runs SPARROW services, interfaces with sensors, and provides local processing.              | Use Raspberry Pi 5 8GB.                                                                       |
| Compute and storage | Compatible Raspberry Pi 5 power supply           | 1   | Required for setup | Bench power supply for flashing, first boot, and troubleshooting before the solar power path is complete.               | Use a power supply compatible with Raspberry Pi 5.                                            |
| Compute and storage | NVMe Base for Raspberry Pi 5                     | 1   | Required           | PCIe expansion board used to connect and boot from an M.2 NVMe SSD.                                                     | Use a Pi 5 compatible NVMe base or equivalent PCIe HAT that fits the enclosure.               |
| Compute and storage | M.2 NVMe SSD, PCIe Gen 3 recommended             | 1   | Required           | Primary storage for OS, local logs, sensor data, and application files.                                                 | Minimum 1TB recommended. Larger capacity may be useful for image and audio heavy deployments. |
| Compute and storage | Raspberry Pi RTC battery case and CR2032 battery | 1   | Required           | Keeps time during power loss so logs, files, and scheduled tasks retain accurate timestamps.                            | Use a Pi 5 compatible RTC battery assembly.                                                   |
| Compute and storage | Pi 3 Click Shield, 40-pin GPIO to mikroBUS       | 1   | Required           | Converts the Pi GPIO header into mikroBUS sockets for Click boards.                                                     | The guide uses socket 1 for Relay 3 Click and socket 2 for Environment 3 Click if installed.  |
| Compute and storage | Witty Pi 5 HAT+                                  | 1   | Required           | Smart power management board for the Raspberry Pi. Provides scheduled on and off behavior and wide-input DC regulation. | Accepts roughly 6V to 30V DC and provides stable 5V to the Pi.                                |
| Power system        | 12V solar panels                                 | 2   | Required           | Solar power generation. Wired in series to create a 24V array.                                                          | Size panels based on deployment load and expected sunlight. The guide shows a 24V array.      |
| Power system        | Victron SmartSolar 100/20 MPPT controller        | 1   | Required           | Charges and maintains the battery bank, manages power profiles, and supplies system voltage.                            | Guide references Victron SmartSolar 100/20 MPPT.                                              |

|                        |                                                                              |                              |                                                  |                                                                                                                                   |                                                                                                                 |
|------------------------|------------------------------------------------------------------------------|------------------------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| Power system           | Victron VE.Direct USB data cable                                             | 1                            | Required                                         | Connects the solar controller to the Raspberry Pi for remote monitoring of voltage, charge state, generation, and battery health. | Guide refers to this as the solar charge controller USB data cable.                                             |
| Power system           | 24V battery bank, LiFePO4 preferred, AGM compatible                          | 1 bank                       | Required                                         | Stores energy for autonomous operation.                                                                                           | Use one 24V battery or two 12V batteries in series. A 24V 100Ah bank is recommended in the specification sheet. |
| Power system           | Battery terminal to SAE connector wire set with ATC fuse holder and 30A fuse | 1                            | Required                                         | Connects battery terminals to the rest of the power wiring with quick disconnect and fused protection.                            | Include spare 30A fuses if the wire set does not include them.                                                  |
| Power system           | SAE connector to PV004 connector cable set with polarity inverter adapter    | 1                            | Required                                         | Connects the battery lead to the solar charge controller wiring path.                                                             | Verify polarity before energizing the system.                                                                   |
| Power system           | Solar panel PV004 extension wire set                                         | 1 set                        | As needed                                        | Connects the solar panel array to the solar charge controller when the panels are placed away from the compute box.               | Length depends on installation layout. Outdoor rated cable is recommended.                                      |
| Networking             | Starlink Mini Kit                                                            | 1                            | Required for satellite build                     | Satellite internet backhaul for remote deployments.                                                                               | Includes Starlink Mini and DC power cable used by this build. Requires Starlink account and active service.     |
| Networking             | Starlink weatherproof Ethernet adapter or connector kit                      | 1                            | Required for satellite build                     | Weatherproofs the Starlink Ethernet connection and protects RJ45 ends from moisture and dirt.                                     | Use with outdoor-rated Ethernet cable.                                                                          |
| Networking             | CAT6A outdoor-rated Ethernet cable                                           | 1                            | Required for satellite build                     | High-speed network link between the Raspberry Pi and Starlink adapter.                                                            | Use UV and weather resistant outdoor cable. Length depends on placement.                                        |
| Networking             | XBee 898/900 RF module                                                       | 1                            | Required for RF build                            | Long-range low-power wireless communication using DigiMesh or point-to-multipoint networking.                                     | Select frequency band appropriate for local regulations.                                                        |
| Networking             | UART to USB adapter for XBee                                                 | 1                            | Required for RF build                            | Connects the XBee RF module to the Raspberry Pi over USB.                                                                         | USB-A or USB-C versions work. The guide uses USB-C in the enclosure example.                                    |
| Networking             | 915 MHz XBee or LoRa antenna                                                 | 1                            | Required for 915 MHz RF build                    | Extends wireless range and improves signal reliability for outdoor deployments.                                                   | Use a weatherproof antenna appropriate for the selected RF module and jurisdiction.                             |
| Networking             | External antenna cable or RP-SMA bulkhead hardware                           | As needed                    | Required when routing antennas outside enclosure | Routes XBee or Wi-Fi antenna connections through the enclosure while supporting a weather-resistant build.                        | Guide references RP-SMA antenna routing and external antenna connectors.                                        |
| Power switching        | Relay 3 Click board                                                          | 1                            | Required                                         | Dual relay board used as a remote on and off switch to manage Starlink power.                                                     | Guide installs Relay 3 Click in socket 1.                                                                       |
| Enclosure and mounting | Large outdoor weatherproof extension cord cover box                          | 1                            | Required for standard build                      | Main weatherproof enclosure for the Raspberry Pi assembly, solar controller, wiring, and connectors.                              | Guide describes large box around 12.5 x 8.5 x 5 inches with multiple cable ports.                               |
| Enclosure and mounting | Dual Lock or black Velcro mounting tape                                      | Several 2 inch pieces        | Required                                         | Secures devices inside the box while allowing removal, repositioning, and vibration reduction.                                    | Use for Pi assembly, XBee module, sensors, and cable management.                                                |
| Enclosure and mounting | Heavy-duty zip ties                                                          | Pack                         | Required                                         | Cable routing, strain relief, attachment to mounts, and field deployment support.                                                 | Use UV-resistant ties for outdoor deployments.                                                                  |
| Wiring and connectors  | 18 to 20 AWG red and black wire spool                                        | 1 spool each color           | Required                                         | Standard wire used throughout the SPARROW unit for low-voltage power connections.                                                 | The guide allows any gauge between 18 and 20 AWG for this build.                                                |
| Wiring and connectors  | 3-port WAGO 221 lever connectors                                             | At least 2, more recommended | Required                                         | Creates safe reusable positive and negative distribution points for load wiring.                                                  | Use separate WAGO connectors for red positive and black negative wires.                                         |
| Wiring and connectors  | Starlink 5.5 x 2.1 mm DC female power socket cable or connector              | 1                            | Required for custom Starlink                     | Provides the female DC connection used to route Starlink power through the relay                                                  | The guide lists both a prebuilt cable and DIY socket connector option.                                          |

|                       |                                                   |        |                           |                                                                                          |                                                                                                      |
|-----------------------|---------------------------------------------------|--------|---------------------------|------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
|                       |                                                   |        | power path                | and box.                                                                                 |                                                                                                      |
| Wiring and connectors | Electrical insulation tape                        | 1 roll | Required                  | Electrical insulation, labeling support, and temporary cable protection during assembly. | Use only as appropriate. Do not rely on tape as a substitute for proper connectors or strain relief. |
| Wiring and connectors | Permanent marker                                  | 1      | Required                  | Cable marking, polarity labeling, and assembly notes.                                    | Use for labeling positive, negative, load, PV, battery, and sensor leads.                            |
| Setup peripherals     | Host PC with internet access                      | 1      | Required                  | Used to flash the Raspberry Pi OS image, download scripts, and configure firmware.       | Windows, macOS, or Linux can be used.                                                                |
| Setup peripherals     | Two available electrical outlets                  | 2      | Required for setup space  | Powers the host PC and Starlink unit in parallel during setup.                           | Additional outlets may be useful for tools and peripherals.                                          |
| Setup peripherals     | M.2 USB caddy                                     | 1      | Required                  | Holds the NVMe SSD while Raspberry Pi Imager writes the operating system.                | USB-C or USB-A caddy is fine if compatible with the host PC.                                         |
| Setup peripherals     | Keyboard and mouse                                | 1 set  | Required for setup        | Used during first boot and local Pi configuration.                                       | Can be omitted after remote access is configured.                                                    |
| Setup peripherals     | Monitor or portable monitor with cable or adapter | 1      | Required for setup        | Used for Raspberry Pi first boot and troubleshooting.                                    | Guide notes HDMI or DisplayPort cable or adapter depending on monitor.                               |
| Setup peripherals     | Mobile phone                                      | 1      | Required for camera setup | Used to configure Reolink cameras and Starlink app workflows.                            | Needed if cameras or Starlink app setup are part of the build.                                       |

## Optional and deployment-specific BOM

| System                  | Component                                                                  | Qty             | Status                                                           | Use in SPARROW                                                                                                                | Sourcing notes                                                                                    |
|-------------------------|----------------------------------------------------------------------------|-----------------|------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Environmental sensing   | Environment 3 Click Board, BME688                                          | 1               | Optional                                                         | Atmospheric sensor for pressure, humidity, temperature, VOCs, VSCs, CO, and hydrogen detection.                               | Guide supports this board out of the box and installs it in socket 2.                             |
| Sensor expansion        | Shuttle Click Expansion Board or BUS Board                                 | As needed       | Optional                                                         | Additional mikroBUS expansion capacity for deployments that need more click boards than the Pi 3 Click Shield alone provides. | The specification sheet mentions Shuttle Click Expansion Board, BUS Board, and Pi 3 Click Shield. |
| Audio sensing           | AudioMoth DEV board                                                        | 1 or more       | Optional                                                         | Acoustic sensing for biodiversity monitoring.                                                                                 | Requires USB microphone firmware if used as USB microphone.                                       |
| Audio sensing           | AudioMoth DEV waterproof case                                              | 1 per AudioMoth | Optional                                                         | Weatherproof enclosure for the AudioMoth DEV board.                                                                           | Guide describes removing the battery cradle and fitting a cable gland.                            |
| Audio sensing           | Micro USB to USB-A cable for AudioMoth                                     | 1 per AudioMoth | Optional                                                         | Connects AudioMoth to the Raspberry Pi.                                                                                       | Average length of at least 3 ft recommended.                                                      |
| Audio sensing           | Nvidia Jetson Orin Nano                                                    | 1               | Only if following the optional AudioMoth materials list verbatim | Listed in the AudioMoth optional section of the guide.                                                                        | Validate whether this is still needed for the current Pi-based reference build before purchasing. |
| Camera sensing          | Reolink Argus Eco Wi-Fi camera                                             | As needed       | Optional                                                         | Wi-Fi camera sensor for image capture and email alert workflow.                                                               | Configure through Reolink app without logging into a Reolink account during setup.                |
| Camera sensing          | Reolink solar panel for camera                                             | 1 per camera    | Optional                                                         | Solar charging accessory for Reolink cameras.                                                                                 | Recommended for long field deployments.                                                           |
| Tough enclosure option  | IP65 or NEMA 4(x) electrical junction box with backplate                   | 1               | Optional replacement                                             | Higher weatherproofing enclosure for rainforest or tough outdoor deployments.                                                 | Use instead of the standard weatherproof extension cord cover box when needed.                    |
| Tough enclosure option  | Assorted waterproof cable glands, 3 to 16 mm, including PG9 and PG19 sizes | Set             | Optional or required for tough box                               | Seals cable entries for power, Ethernet, AudioMoth USB, antenna, and sensor cables.                                           | Guide references 25 mm holes for PG19 and 16 mm holes for PG9 antenna routing.                    |
| Tough enclosure option  | Outdoor-rated conduit                                                      | As needed       | Optional but recommended                                         | Protects exposed wiring from wildlife, moisture, physical damage, and UV exposure.                                            | Select conduit compliant with local electrical code.                                              |
| Mounting and deployment | Solar panel mounting brackets                                              | As needed       | Deployment dependent                                             | Holds panels at the proper angle and elevation.                                                                               | Required for open-area deployments and other ground-mounted configurations.                       |

|                         |                                          |           |                           |                                                                               |                                                                             |
|-------------------------|------------------------------------------|-----------|---------------------------|-------------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Mounting and deployment | Pole support for Starlink antenna        | As needed | Deployment dependent      | Positions Starlink antenna with clear sky view.                               | Use appropriate anchoring and guy wires if exposed to wind.                 |
| Mounting and deployment | Guy wires and anchors                    | As needed | Deployment dependent      | Stabilizes poles and panel mounts in open or windy sites.                     | Match hardware to local wind and soil conditions.                           |
| Wireless accessories    | 6dB plastic Wi-Fi antennas               | As needed | Optional                  | External Wi-Fi antenna option referenced in the tough enclosure instructions. | Requires enclosure drilling and RP-SMA connector mounting if used.          |
| Wireless accessories    | 10dB long-range fiberglass XBee antenna  | 1         | Optional upgrade          | Higher-gain external antenna option for longer range XBee deployments.        | Guide references additional drilling and PG9 glands for this antenna route. |
| Battery protection      | Weatherproof battery tote or bin         | 1         | Recommended for field use | Protects the battery bank from moisture and environmental exposure.           | Add cable glands, elevation, and insulation depending on site conditions.   |
| Battery protection      | Thermal insulation for battery enclosure | As needed | Cold climate option       | Helps maintain battery temperature in cold deployments.                       | Use only materials appropriate for battery safety and ventilation needs.    |

## Tools and shop supplies

| Tool or supply                                          | Use                                                                                                |
|---------------------------------------------------------|----------------------------------------------------------------------------------------------------|
| Electric drill with conical step bit set, 4 mm to 35 mm | Drills cable gland and routing holes when needed.                                                  |
| Basic hand tools                                        | Pliers, scissors, angle or micro cutting pliers, wire strippers, and precision screwdriver set.    |
| Small Phillips screwdriver, #0 or #1                    | Used for Pi, NVMe base, HAT, and small board hardware.                                             |
| Anti-static wrist strap                                 | Optional but recommended for handling boards and SSD.                                              |
| Magnifying glass and plastic spudger                    | Useful for small connectors, ribbon cables, and safe handling.                                     |
| Protective eyewear and gloves                           | Recommended personal protective equipment for assembly and field installation.                     |
| Fire safety item                                        | Fire blanket, container of sand, or appropriate extinguisher nearby when working with batteries.   |
| Precision knife, such as X-Acto knife                   | Used only if an AudioMoth micro USB connector needs careful trimming to fit through a cable gland. |

## Software, accounts, and setup dependencies

| Dependency                                          | Purpose                                                                                       |
|-----------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Raspberry Pi Imager                                 | Used to flash Raspberry Pi OS to the M.2 SSD.                                                 |
| Raspberry Pi OS, 64-bit                             | Operating system selected in the guide.                                                       |
| SPARROW setup script                                | Available from the SPARROW GitHub repository. Used to configure the Raspberry Pi environment. |
| SPARROW account or self-hosted dashboard account    | Needed to obtain a device access key if using SPARROW Studio.                                 |
| AudioMoth Flash App                                 | Required only when using AudioMoth and flashing USB microphone firmware.                      |
| Reolink mobile app                                  | Required only when configuring Reolink cameras.                                               |
| Starlink account and active service                 | Required when using Starlink Mini for connectivity.                                           |
| Internet connection for downloads and Docker images | Required during software setup and firmware download steps.                                   |

## Procurement and substitution notes

- Confirm local electrical, radio, environmental, and installation compliance before purchasing parts.
- Select RF modules, antennas, and frequency bands that are legal for the deployment country.
- Size solar panels and battery capacity to the site, sensor load, network duty cycle, and expected weather.
- Use outdoor-rated, UV-resistant cable, conduit, glands, fasteners, and enclosures for exposed installations.
- Keep positive and negative conductors physically separate until they are connected to the correct terminals.
- Verify all polarities before connecting the battery and solar panels to the solar charge controller.
- Plan cable lengths before cutting. Leave slack for routing, strain relief, and service access.
- Purchase spare fuses, WAGO connectors, cable glands, zip ties, and mounting tape for field repairs.

## Safety and compliance notice

### **Builder responsibility**

SPARROW is an open-source, end-user assembled system. Builders are responsible for selecting compliant components, following manufacturer instructions, using proper wiring practices, and ensuring the completed device meets local electrical, radio, environmental, and installation requirements.